Sudarsan Srinivasan

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Summary

Mechanical engineer with 3+ years developing powertrain components across concept design, prototyping, validation, and production support, with additional experience in consumer products, industrial equipment, and production automation. Skilled in 3D CAD design, GD&T, tolerance analysis, DFM, supplier development, and cross-functional integration.

EXPERIENCE

Stealth Startup **Sep 2026 - Present**
Founding Hardware Engineer *San Francisco Bay Area*

- Lead development of an automated production process for precision electromechanical assemblies, defining equipment requirements, mechanical concepts, tooling options, and manufacturing workflows.
- Evaluate surface-preparation methods and coating options for adhesive-bonded components; design DOEs and adhesion-strength tests to assess process parameters and bonding reliability.

SharkNinja **Jan 2026 - Jun 2026**
Mechanical Engineering Co-op, New Product Development *Needham, MA*

- Owned design simplification and risk assessment of slush machine auger against KPIs across eight recipes; confirmed design parity on 75% of test cases. Proposed and validated condenser-fan control changes that brought the remaining recipes within KPIs, supporting production release.
- Investigated compressor stalls across five units through failure reproduction, flow-path isolation, and teardown; documented shaft seizure and witness marks, coordinated supplier investigation, and supported software containment.
- Built a data analysis tool for processing 73-column raw data logs from frozen treats machines: automated data cleaning, anomaly flagging, and chart generation, cutting analysis time from 15 minutes to under 10 seconds per run

The Boring Company **Sep 2025 - Dec 2025**
Integration Engineering Intern *Bastrop, TX*

- Led subsystem integration for the MB-1 TBM, coordinating across teams to design and integrate ventilation, fluid distribution, and gantry systems; resolved build-blocking issues including part kitting errors, routing conflicts, and supply delays to keep builds on schedule
- Owned head-section and washbox integration covering 2,000+ parts within the 10,000+ part VC-6 Vertical Cassette assembly; coordinated build readiness, instructions, and discrepancy resolution, helping keep the assembly on schedule.

Bajaj Auto Limited **Jan 2021 - Jul 2024**
Mechanical Design Engineer, Powertrain Design R&D *Pune, IN*

- Designed die-cast aluminum cases and injection-molded covers for ICE and EV powertrains, applying GD&T, tolerance analysis, and DFM; collaborated with CAD designers on NX models and drawings through validation and release support.
- Proposed and implemented clutch-cover wall and rib changes, reducing mass by 110 g (11%); performed sealing tests and coordinated CAD, CAE, NVH, and durability validation through production release.
- Designed paper and metal gaskets from mating-interface requirements through prototype validation; defined geometry and materials, reviewed supplier drawings, and used ANSYS contact-pressure FEA and pressure-film testing to refine designs.
- Automated gasket modeling and mating-profile checks in NXOpen, generating comparison tables from sealing faces and reducing task duration by 76%.

PROJECT EXPERIENCE

Autonomous Cleaning Robot **Jan 2025 - Dec 2025**
Boeing Advanced Research Collaboration / University of Washington *Seattle, WA*

- Led chassis design, CAD packaging, and prototype integration for an industrial-sump cleaning robot, coordinating the drivetrain, cleaning system, battery, and enclosures; demonstrated autonomous movement with the initial prototype.
- Generated multiple cleaning system concepts and built test fixtures to compare cleaning effectiveness on an inclined epoxy surface, using a Pugh matrix to down-select the most feasible solution for wet, low-maintenance operation.

TECHNICAL SKILLS

- **CAD and PLM**: Siemens NX, SolidWorks (CSWP), Siemens Teamcenter
- **Mechanical design**: GD&T, Tolerance stack-up analysis, DFM/DFA, DFMEA, Design Validation Planning (DVP), Electromechanical Integration
- **Analysis and testing**: ANSYS Mechanical, Structural analysis, Root cause analysis, DOE, FEA
- **Manufacturing**: Die casting, Injection molding, Sheet metal, 3D printing, Prototyping, Supplier development, High volume manufacturing, Mass production design
- **Programming**: NXOpen, Python, MATLAB

EDUCATION

University of Washington **Sep 2024 - Aug 2026**
MS, Mechanical Engineering *Seattle, WA*

National Institute of Technology, Trichy **Jun 2016 - Jun 2020**
B Tech, Mechanical Engineering *Trichy, IN*